

Central Asian University

INTERNSHIP REPORT

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Student ID:	230514
Programme:	Computer Science / IT
Company:	Alif Transportation LLC
Department:	IT & Data Operations
Duration:	01 March 2026 – 30 April 2026 (8 weeks)

1. Summary

I did my internship at Alif Transportation LLC from March to April 2026, working in the IT and Data Operations department for 8 weeks. The main thing I worked on was a QR code and barcode detection program that I built in Python using OpenCV. This was also my project for the Image Processing module resit exam. The program ended up working well — 94% accuracy on my test dataset and it runs in real time at 27 frames per second. The internship was a good experience overall and I learned a lot both technically and professionally.

2. About the Company

Alif Transportation is a trucking and logistics company that operates fleets of commercial vehicles across the US under the Alif, TTL and DT carrier names. They have a large number of drivers, dispatchers and support staff. The IT team I worked with handles driver data, compliance documents, and automation tools mostly written in Python.

It turned out to be a good match for my project because the team already uses Python for data processing and they actually scan driver documents regularly, so a QR/barcode detector is something they could realistically use.

3. What I Was Trying to Do

My goals for the internship were:

- Build a working QR code detection system and connect it to something real.
- Practice writing proper Python code with comments and documentation.
- Learn how to use Git and GitHub properly for version control.
- Get some experience working in a real office environment.

4. What I Did Each Week

Weeks 1–2: Set up the development environment, collected 50 test images, and read about how QR detection works. Also got familiar with how the company operates.

Weeks 3–4: Wrote the main detection code. Added webcam support and the batch evaluation mode. Fixed a bug with empty detections crashing the program.

Week 5: Tested everything on the full dataset and calculated the metrics. 94% accuracy, 0 false positives, 27 FPS.

Weeks 6–7: Wrote the technical report and made the presentation slides. Did a practice run in front of colleagues.

Week 8: Final cleanup, submitted everything, exit meeting with supervisor.

5. What I Learned

Technical	Professional
Python and OpenCV in practice	Writing clearly for a real audience
Git and version control	Managing time over a long project
Evaluation metrics and testing	Explaining technical work simply
Real-time video processing	Working with a team

6. Difficulties I Ran Into

Problem: The program crashed when a code was detected but the decoder returned empty string.

How I solved it: Added a simple check before processing: if data is empty, skip it.

Problem: Some real-world photos were hard for the detector.

How I solved it: Documented the failure cases and listed WeChatQRCode as a future improvement.

Problem: Understanding the company data privacy rules at the start.

How I solved it: Read the policy document carefully and only used anonymised test images throughout.

7. Conclusion

The 8 weeks went by faster than I expected. I came in not really knowing how to use OpenCV and left with a working project I am actually proud of. The company was helpful and the environment was professional but not stressful. I think this internship made me more confident about my programming skills and gave me a better idea of what working in an IT department is actually like.

If I had to do it again I would start collecting the dataset earlier — that took more time than I expected. But overall it was a good experience.